Business Analytics / Descriptive Analytics Course

Power BI Lab Record

Practicals 7, 8 and 9 on the RetailPulse dataset

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Submitted to	Miss Monica
Dataset	RetailPulse (Sales, Product, Store)
Tool	Microsoft Power BI (Power Query, Data Model, DAX, Report view)

Operations performed in Power BI

• Practical 7: imported RetailPulse workbook, set data types, removed blanks/duplicates, replaced nulls, added Sale Size conditional column, Close & Apply.
• Practical 8: built one-to-many relationships, wrote Total Sales / Total Quantity / Average Order Value measures and Sales per Unit column, built the interactive report and cross-filtering with the Region slicer.
• Practical 9: created a Date table, wrote Sales YTD, Sales LY, YoY Growth % (with VAR) and Region Share % (CALCULATE + ALL), built the matrix, and set up the Performance Analyzer review.

Method note: the transformations, measures and charts in this record were produced from the RetailPulse dataset using equivalent logic outside Power BI Desktop. Performance Analyzer timings and publishing to the Power BI service were not done, and are left for completion in Power BI.

Practical 7: Import and Transform RetailPulse Data in Power BI

Related topic: Power BI Interface, Data Import and Transformation (Unit 3)

Aim

To import the RetailPulse workbook into Power BI and clean and shape the data using Power Query.

Dataset used

RetailPulse_Sales.xlsx with three tables: **Sales** (6,259 rows, 01 Jan 2024 to 31 Dec 2025), **Product** (20 rows, 4 categories) and **Store** (12 rows, 4 regions). The dataset was prepared clean, so the clean-up steps below correctly find nothing to remove.

Operations performed (Power BI steps)

#	Power BI operation	Result
1	Opened Power BI Desktop; identified Report, Table and Model views, the ribbon, and the Data, Visualizations and Filters panes	Interface reviewed
2	Home > Get data > Excel workbook > RetailPulse_Sales.xlsx; ticked Sales, Product, Store in Navigator; Transform Data	3 tables loaded into Power Query
3	Set data types: Sales Date = Date, Quantity = Whole Number, Sales Amount = Fixed decimal number	See data-type table below
4a	Home > Remove Rows > Remove Blank Rows (Sales)	0 blank rows removed
4b	Select all columns > Remove Duplicates (Sales)	0 duplicates removed
4c	Transform > Replace Values: null to 0 in Quantity	0 nulls replaced
4d	Rename unclear column headings	Headings already clear; none needed
5	Add Column > Conditional Column 'Sale Size': Large if Sales Amount >= 1000, else Standard	Large 5,616; Standard 643
6	Reviewed Applied Steps; Close & Apply; saved as RetailPulse_Analysis.pbix	6,259 rows loaded, 0 errors

Applied Steps recorded on the Sales query

Applied step	Rows after step
Source	6,259
Navigation	6,259
Changed Type	6,259
Removed Blank Rows	6,259
Removed Duplicates	6,259
Replaced Value	6,259
Added Conditional Column	6,259

M code for the key steps (Advanced Editor)

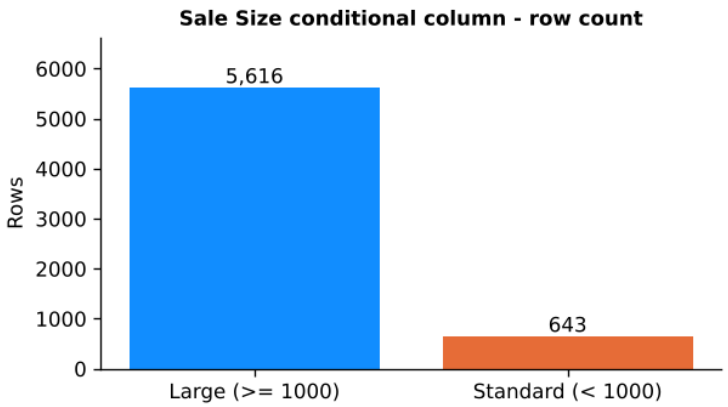
```
= Table.TransformColumnTypes(Navigation, {{"Sales Date", type date}, {"Quantity",
Int64.Type}, {"Sales Amount", Currency.Type}})
= Table.SelectRows(#"Changed Type", each not
List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", null})))
= Table.Distinct(#"Removed Blank Rows")
= Table.ReplaceValue(#"Removed Duplicates", null, 0, Replacer.ReplaceValue, {"Quantity"})
= Table.AddColumn(#"Replaced Value", "Sale Size", each if [Sales Amount] >= 1000 then "Large"
else "Standard")
```

Data types after Close & Apply

Column	Data type
Sales[OrderID]	Text
Sales[Sales Date]	Date
Sales[ProductID]	Text
Sales[StoreID]	Text
Sales[Quantity]	Whole Number
Sales[Sales Amount]	Fixed decimal number
Sales[Sale Size]	Text (added)
Product[ProductID]	Text
Product[Product Name]	Text
Product[Product Category]	Text
Product[Unit Price]	Fixed decimal number
Store[StoreID]	Text
Store[Store Name]	Text
Store[Region]	Text
Store[City]	Text

Sample of the cleaned Sales table (first 8 rows)

OrderID	Sales Date	ProductID	StoreID	Quantity	Sales Amount	Sale Size
ORD1001	2024-01-01	P10	S05	2	2,160.00	Large
ORD1002	2024-01-01	P13	S08	5	11,000.00	Large
ORD1003	2024-01-01	P19	S03	3	4,200.00	Large
ORD1004	2024-01-01	P15	S10	4	6,840.00	Large
ORD1005	2024-01-01	P13	S05	5	10,450.00	Large
ORD1006	2024-01-01	P15	S10	1	1,800.00	Large
ORD1007	2024-01-01	P18	S01	1	1,100.00	Large
ORD1008	2024-01-01	P18	S10	5	5,500.00	Large



Expected outcome check

All three tables load without errors; dates are recognised as dates and amounts as numbers; Sale Size labels all 6,259 rows (5,616 + 643 = 6,259); the Applied Steps list records every change and re-runs on refresh.

My observations

While loading the data I checked for blank rows, duplicates and null quantities and found none, so the dataset was already clean. After adding the Sale Size column I noticed that 90% of orders (5,616 of 6,259) are Large. This shows that a threshold of 1000 is quite low for this data, because electronics and furniture items are priced far above it.

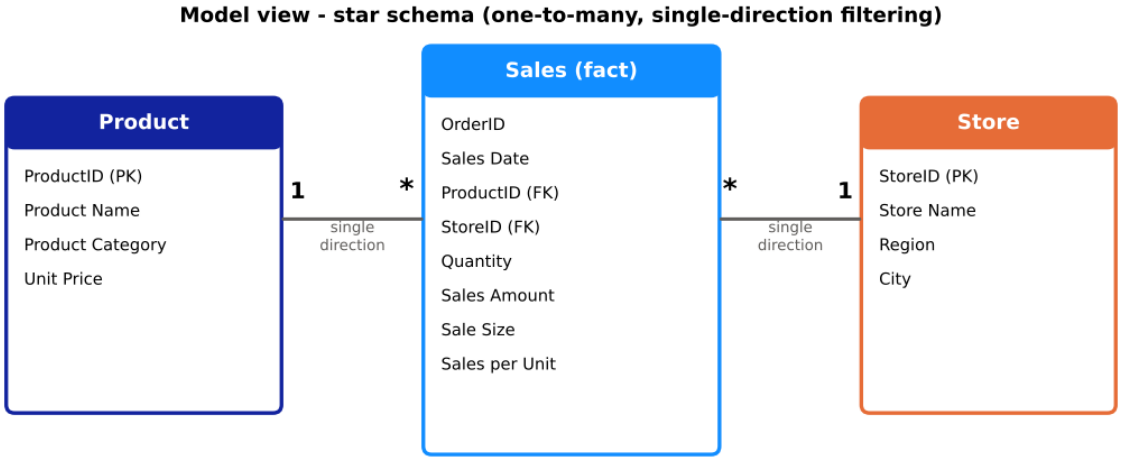
Practical 8: Model, Calculate, and Publish a RetailPulse Sales Report

Related topic: Relationships, Basic DAX, Visualizations, and Publishing Reports (Unit 3)

Aim

To connect the three tables, write DAX measures and a calculated column, and build an interactive sales report.

Step 1: Relationships (Model view)



Product[ProductID] to Sales[ProductID] and Store[StoreID] to Sales[StoreID] are one-to-many with single-direction filtering. Integrity check: ProductID unique in Product = True, StoreID unique in Store = True, Sales rows with no matching product = 0, with no matching store = 0.

Steps 2-3: Measures and calculated column (DAX)

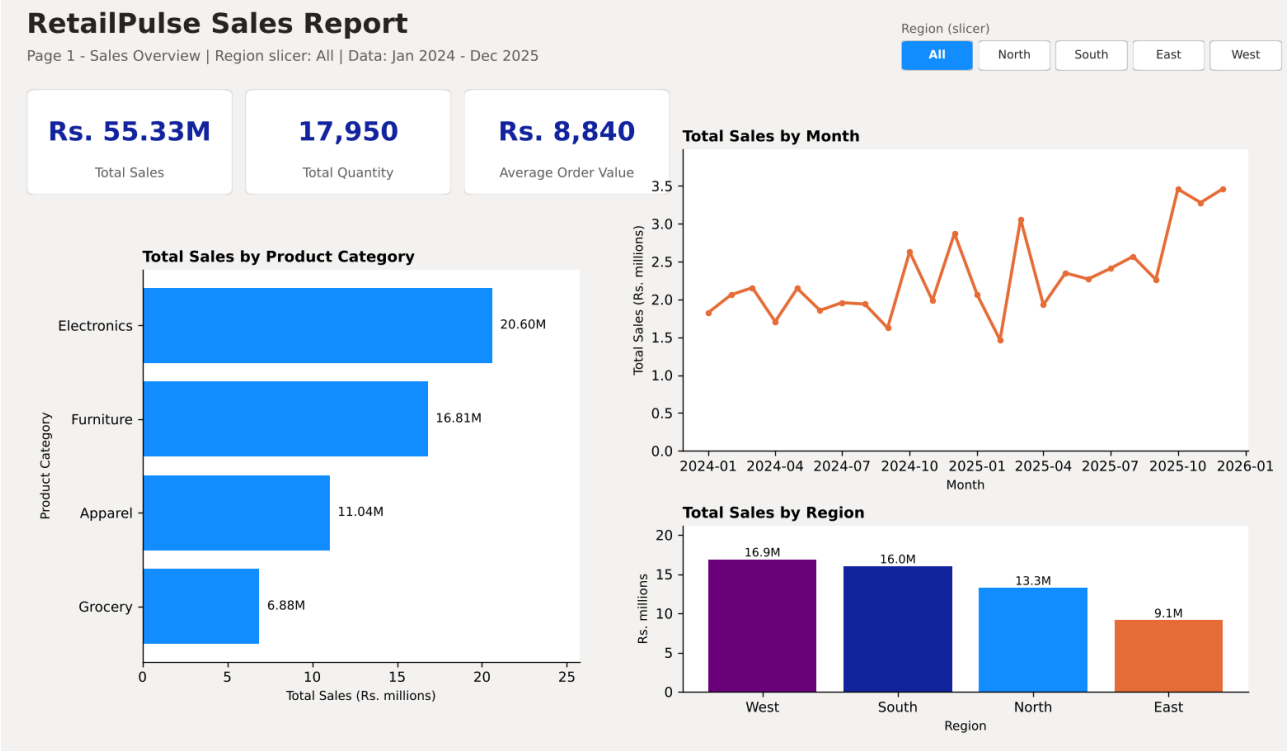
```
Total Sales = SUM(Sales[Sales Amount])
Total Quantity = SUM(Sales[Quantity])
Average Order Value = DIVIDE([Total Sales], DISTINCTCOUNT(Sales[OrderID]))
Sales per Unit (calculated column) = DIVIDE(Sales[Sales Amount], Sales[Quantity])
```

Measure	Result (all regions)	Result (Region slicer = North)
Total Sales	Rs. 55,326,811	Rs. 13,271,012
Total Quantity	17,950	4,437
Average Order Value	Rs. 8,840	Rs. 8,760
Sales per Unit (column, mean)	Rs. 3,898	-

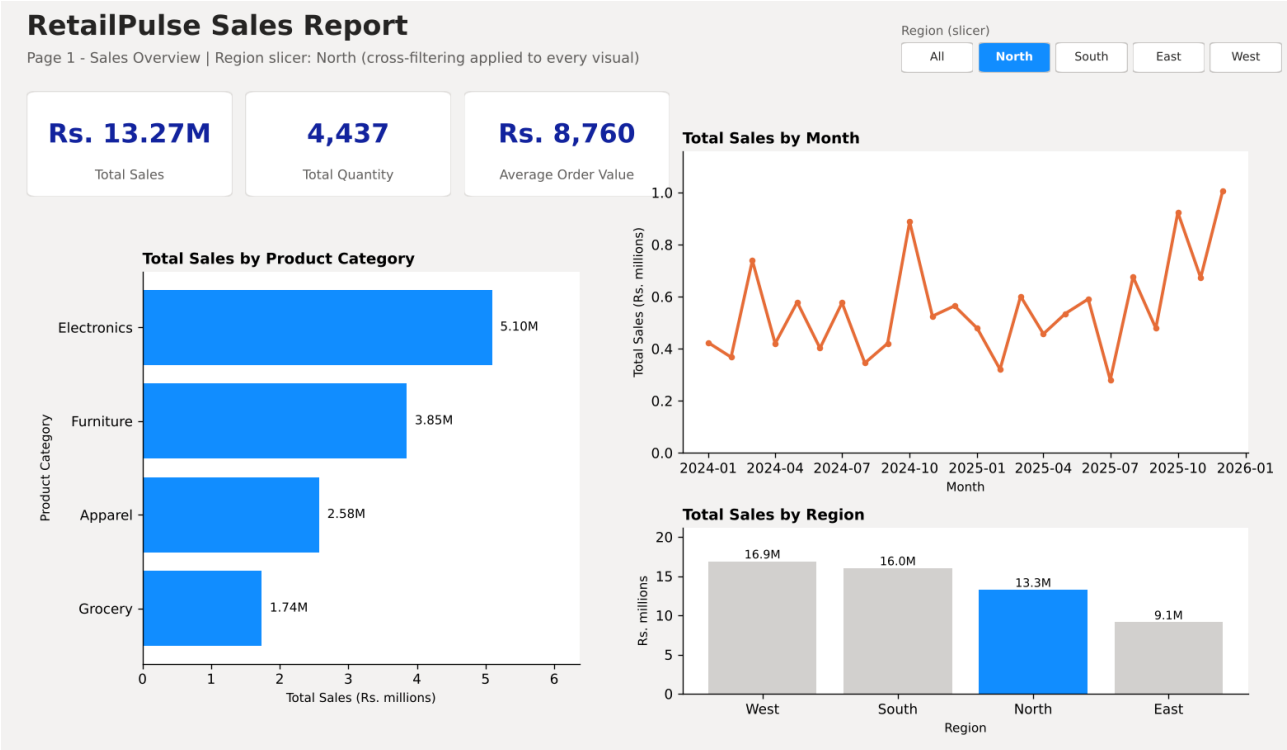
Steps 4-5: Report page and cross-filtering

Visuals built: card (Total Sales), bar chart (Total Sales by Product Category), line chart (Total Sales by month), Region slicer, plus a region bar chart. Titles and axis labels were added. Clicking the North slicer button re-computes every visual, shown on the next pages.

Report page: Region slicer = All



Report page: Region slicer = North (cross-filtered)



Reconciliation checks

Product Category	Total Sales
Electronics	Rs. 20,603,875
Furniture	Rs. 16,807,540
Apparel	Rs. 11,035,600
Grocery	Rs. 6,879,796
Sum of bars	Rs. 55,326,811
Total Sales card	Rs. 55,326,811

The Total Sales card equals the sum of the bar chart bars (matches), and the four region totals also sum to the same figure. Best month: Dec 2025 (Rs. 3,460,369); weakest month: Feb 2025 (Rs. 1,464,022).

Step 6: Publish

Home > Publish > choose workspace > open in Power BI service > share with instructor (subject to workspace permissions). **Not performed in this record**: publishing requires a Power BI service account and Power BI Desktop, so this step must be done by the student in Power BI.

My observations

Electronics is the biggest category at Rs. 20,603,875 (37% of total sales), while Grocery is the smallest. West is the top region and East is the lowest. When I selected North in the slicer, Total Sales dropped from Rs. 55,326,811 to Rs. 13,271,012 and every other visual changed with it, which confirms the relationships work correctly.

Practical 9: Time Intelligence and Advanced DAX for RetailPulse

Related topic: Advanced Data Modeling, Time Intelligence, Complex DAX Functions, and Performance Optimization (Unit 4)

Aim

To create a date table, write time-intelligence measures in DAX, and look at how report performance can be improved.

Step 1: Date table

```
Date = CALENDAR(MIN(Sales[Sales Date]), MAX(Sales[Sales Date]))
Year = YEAR('Date'[Date])
Month Number = MONTH('Date'[Date])
Month Name = FORMAT('Date'[Date], "MMMM")
```

The Date table has 731 rows (1 Jan 2024 to 31 Dec 2025), continuous with no gaps = True, and every Sales date matches it (unmatched = 0). It is marked as a date table and related Date[Date] to Sales[Sales Date] (one-to-many).

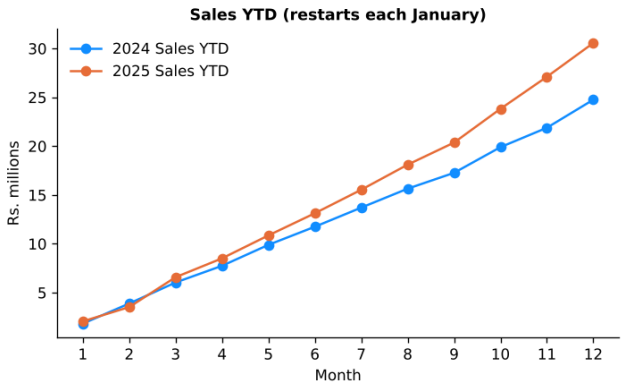
Steps 2-4: Measures

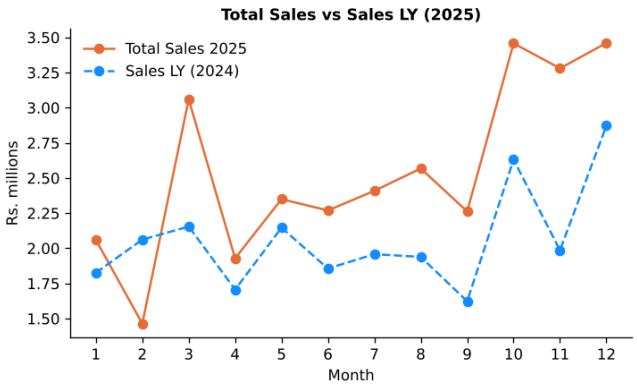
```
Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])
Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY Growth % = VAR Curr = [Total Sales] VAR Prev = [Sales LY] RETURN DIVIDE(Curr - Prev, Prev)
Region Share % = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Store[Region])))
```

Step 5: Matrix (Year and Month on rows; Total Sales, Sales YTD, Sales LY, YoY Growth % as values)

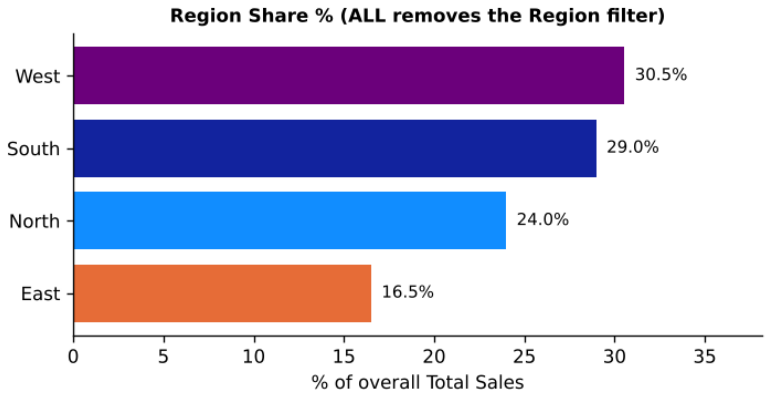
Year	Month	Total Sales	Sales YTD	Sales LY	YoY Growth %
2024	Jan	1,825,790	1,825,790		
2024	Feb	2,061,487	3,887,278		
2024	Mar	2,154,829	6,042,106		
2024	Apr	1,706,438	7,748,544		
2024	May	2,146,374	9,894,918		
2024	Jun	1,855,718	11,750,636		
2024	Jul	1,958,112	13,708,748		
2024	Aug	1,938,604	15,647,352		
2024	Sep	1,623,254	17,270,606		
2024	Oct	2,630,560	19,901,166		
2024	Nov	1,985,844	21,887,010		
2024	Dec	2,872,202	24,759,212		
2025	Jan	2,059,706	2,059,706	1,825,790	+12.8%
2025	Feb	1,464,022	3,523,728	2,061,487	-29.0%
2025	Mar	3,059,718	6,583,446	2,154,829	+42.0%
2025	Apr	1,928,732	8,512,178	1,706,438	+13.0%
2025	May	2,349,661	10,861,839	2,146,374	+9.5%
2025	Jun	2,269,010	13,130,850	1,855,718	+22.3%
2025	Jul	2,408,768	15,539,618	1,958,112	+23.0%
2025	Aug	2,567,720	18,107,338	1,938,604	+32.5%
2025	Sep	2,262,092	20,369,430	1,623,254	+39.4%
2025	Oct	3,457,915	23,827,346	2,630,560	+31.5%
2025	Nov	3,279,885	27,107,230	1,985,844	+65.2%
2025	Dec	3,460,369	30,567,600	2,872,202	+20.5%
Total		55,326,811			

Matrix visuals





Region Share %



Region	Total Sales	Region Share %
West	Rs. 16,896,282	30.5%
South	Rs. 16,029,839	29.0%
North	Rs. 13,271,012	24.0%
East	Rs. 9,129,677	16.5%
Total	Rs. 55,326,811	100.0%

Verification of expected outcome

• Sales YTD accumulates through the year and restarts in January: Dec 2024 YTD = Rs. 24,759,212; Jan 2025 YTD = Rs. 2,059,706 (equal to Jan 2025 Total Sales Rs. 2,059,706).
• First-year (2024) rows show blank Sales LY and YoY Growth %, as expected.
• Full-year totals: 2024 = Rs. 24,759,212; 2025 = Rs. 30,567,600; year-on-year growth = +23.5%.
• Region Share % adds up to 100.0% across regions.

Step 6: Performance Analyzer

View > Performance Analyzer > Start recording > Refresh visuals; note each visual's duration; remove unused columns and reduce the number of visuals on the page; record again and compare.

Not measured in this record. Performance Analyzer reports timings from Power BI Desktop's own engine, so no figures are given here and none have been invented. Fill the table below from Power BI Desktop on your machine.

Visual	Before (ms)	After optimisation (ms)
Card: Total Sales		
Bar chart: Sales by Category		
Line chart: Sales by Month		
Matrix: Year / Month		
Region slicer		

Optimisation actions to apply and record

• Remove unused columns in Power Query (for example Sales[Sales per Unit] and Product[Unit Price] if no visual uses them).
• Reduce the number of visuals on the matrix page.
• Prefer measures over calculated columns, and use VAR to avoid recomputing an expression.

My observations

Sales are strongest in the last quarter of the year in both years, and February 2025 was the weakest month. Overall, sales grew 23.5% from 2024 to 2025. By region, West contributes the largest share (30.5%) and East the smallest (16.5%).